

This by no means implies or excludes similar questions or marks distribution on the exam.

1. (8 marks) Use the rules of inference to show that the following argument is valid. Provide the rule for each step.

	STEPS	REASON
p	1)	
$q \rightarrow r$		
$(p \wedge q) \rightarrow (s \vee \neg r)$	2)	
q	3)	
$\therefore \neg s \rightarrow u$	4)	
	5)	
	6)	
	7)	
	8)	
	9)	
	10)	
	11)	
	12)	
	13)	
	14)	

3. [16 marks] The System Administrator has set the following rules for the password:

- ✓ The password is a string made of 8 characters.
- ✓ The available characters are 26 upper case letters of English alphabet A, B, C... Z and 10 digits 0, 1... 9.

- a) [2 marks] How many passwords have letters only?
- b) [2 marks] How many passwords with letters only have no repeated letters?
- c) [3 marks] How many passwords have five letters "C", two letters "B" and a digit?
- d) [3 marks] The password is considered weak if it contains letters only or digits only. How many passwords have at least one digit and at least one letter?
- e) [3 marks] How many passwords with digits only have non-decreasing sequence of digits and start with 5? The sequence of eight digits $D_1D_2D_3D_4D_5D_6D_7D_8$ is non-decreasing if $D_1 \leq D_2 \leq D_3 \leq D_4 \leq D_5 \leq D_6 \leq D_7 \leq D_8$.
- f) [3 marks] How many passwords have three letters "A", two letters "B", two letters "C" and one letter "D" such that no A's are adjacent.

4. [8 marks] Circle one answer only; you MUST justify the answer. No marks will be given for answers without justification. No marks will be given when justification does not imply the correctness of the answer.

Sets A and B are defined as follows: $A = \{0, 1, 2, 3\}$ and $B = \{-3, -2, -1\}$

a) $\exists x \in A, \exists y \in A$ such that $x + y = 5$ TRUE/FALSE

b) $\forall x \in A, \forall y \in B$, then $x \cdot y < 0$ TRUE/FALSE

c) $\exists x \in A, \forall y \in B$ such that $x \cdot y \in A$ TRUE/FALSE

d) $\forall x \in A, \exists y \in B$ such that $\{x - y\} \subseteq \{-29, 0, 1, x + y\}$ TRUE/FALSE

- 5a. (2 marks) Negate and simplify the following statement:

$$\forall x [p(x) \rightarrow (\neg r(x))]$$

- 5b. (2 marks) Write the negation of the following statement:

If [c divides a $\wedge$ c does not divide b] then c does not divide (a+ b).

6. (7 marks) Consider the following programming segment:

***You should note that the two *for loops* are *nested loops*.

```
counter = 150

for i = 3 to n do {

    counter = counter + 5

    for j = i+1 to 5n+3 do {

        counter = counter + 10

    }

}
```

Determine the value that the variable *counter* will have after the segment is executed. Provide your answer in terms of n. ***In order to receive marks your approach must be combinatorial - which means that you have to consider combinatorial structures like combinations, permutations or arrangements. Answers that rely on sigma notation and/or solving summation will not be accepted.***

7. (4 marks) What is the probability that a randomly selected binary string of length 16 starts with 0111 or does NOT end with 101 exclusively (“exclusive or” = one or the other but not the both)?